Perverse Sheaves

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1 Introduction

1.1 Overview of the Course

X a smooth complex projective algebraic variety of dimension $\dim_{\mathbb{C}} X = d$. Then there is an isomorphism $H^i(X,\mathbb{Q}) \to H^{2d-i}(X,\mathbb{Q})$ given by multiplication with some $l \in H^2(X,\mathbb{Q})$.

$$H^i(X,\mathbb{Q}) \xrightarrow{l} H^{i+2}(X,\mathbb{Q}) \xrightarrow{l} \cdots \xrightarrow{l} H^{2d-i}(X,\mathbb{Q})$$
$$\sim$$

We then get $b_i \leq b_{i+2}$ if $i < d$ and $b_i \geq b_{i+2}$ if $i > d$.

$$b_0 \leq b_2 \leq \cdots \leq b_{\text{middle}} \geq \cdots \geq b_{2d},$$
$$b_1 \leq b_3 \leq \cdots \leq b_{\text{middle}'} \geq \cdots \geq b_{2d-1}.$$

Definition 1.1.1. *Let X be a complex projective variety. Then we define,*

$$H^{ij}(X) := H^i(X, \Omega_X^j).$$

Theorem 1.1.1.

$$H^k(X,\mathbb{C}) = \bigoplus_{i+j=k} H^{i,j}(X),\ H^{i,j} \cong \overline{H^{j,i}(X)}.$$

Furthermore,

$$b_k = \sum_{i+k} b_{i,j}, b_{i,j} = b_{j,i}.$$

Corollary 1.1.1. *if k is odd then b_k is even.*

Theorem 1.1.2 (Deligne)**.** *Let $f : X \to Y$ be a smooth morphism between smooth projective complex varieties. Then we have,*

$$Rf_*\underline{\mathbb{Q}}_X \cong \bigoplus_{i\in\mathbb{Z}} \mathcal{H}^i(Rf_*\underline{\mathbb{Q}}_X)[-i],$$

where $\underline{\mathbb{Q}}_X$ denotes the constant $\mathbb{Q}$-sheaf on X. .

Remark 1.1.1. *Consider,*

$$S^1 \to S^3 \xrightarrow{f\ Hopf} S^2$$

Then $Rf_*\underline{\mathbb{Q}}_{S^3}$. However, $\mathcal{H}^0(Rf_*\underline{\mathbb{Q}}_{S^3}) = \underline{\mathbb{Q}}_{S^2}$ and $\mathcal{H}^1(Rf_*\underline{\mathbb{Q}}_{S^3}) = \mathbb{Q}_{S^2}$. Therefore,

$$Rf_*\underline{\mathbb{Q}}_{S^3} \ncong \underline{\mathbb{Q}}_{S^2} \oplus \underline{\mathbb{Q}}_{S^2}[-1], H^*(S^3) \ncong H^*(S^2) \oplus H^{*-1}(S^2).$$

Hence f does not satisfy Theorem 1.1.2

The starting point of the theory was that over which spaces, do we have poincare duality. This question led to the invention of intersection cohomologies.

Definition 1.1.2 (Intersection Cohomology)**.** *For a variety X over $\mathbb{C}$,*

There is a perverse sheaf $\mathrm{IC}_X \in D^b_c(X)$ in the derived category of constructible sheaves on X such that

$$\mathrm{IH}^*(X, \mathbb{Q}) = H^*(X, \mathrm{IC}_X)$$

which is the cohomology of this complex of sheaves. Perverse sheaves, Belinson-Bernstein-Deligne 1983, BBD + Gaber also extended Deligne's theorem to $f : X \to Y$ proper.

2 Recollection of Preliminaries

Let X be a topological space and k be a commutative ring. The category of presheaves of k-modules over X is denoted by $\mathrm{PreShv}(X; k)$, similarly the sheaves of k-modules over X is denoted by $\mathrm{Shv}(X; k)$.

Example 2.0.1.

1. *For all $M \in \mathrm{Mod}(k)$, there is a constant sheaf $\underline{M}_X \in \mathrm{Shv}(X; k)$.*
2. *For $r > 0$, consider morphisms of groups,*

$$p : \pi_1(X, x_0) \to \mathrm{Aut}(k^{\oplus r}).$$

Define $L_p 4$ as follows.

Example 2.0.2. *For $r > 0$, consider morphisms of groups,*

$$p : \pi_1(X, x_0) \to \mathrm{Aut}(k^{\oplus r}).$$

Define $L_p 4$ as follows. Let $\pi : \tilde{X} \to X$ be the universal covering space. For $U \subseteq X$,

$$L_p := \{\pi^{-1}(U) \to k \mid \forall g \in \pi_1(X, x_0), f(x \cdot g) = p(g)^{-1} f(x)\}.$$

This is called a local sysmte L_p (locally trivial).

Theorem 2.0.1. $\mathrm{Shv}(X; k)$ *is an abelian category.*

Proposition 2.0.1. *The canonical inclusion $\mathrm{Shv}(X; k) \to \mathrm{PreShv}(X; k)$ admits a left adjoint and such functor is called the sheafification,*

$$\mathrm{PreShv}(X; k) \ni \mathcal{F} \mapsto \mathcal{F}^\dagger \in \mathrm{Shv}(X; k).$$

Explicitly, for $\mathcal{F} \in \mathrm{PreShv}(X; k)$ and $\mathcal{G} \in \mathrm{Shv}(X; k)$, we have

$$\mathrm{Mor}_{\mathrm{PreShv}}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{G}),$$

which is natural. In particular, setting $\mathcal{G} = \mathcal{F}^\dagger$, we obtain a natural map $\mathcal{F} \to \mathcal{F}^\dagger$ which is obtained from $\mathrm{id} \in \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{F}^\dagger)$.

Proposition 2.0.2. *The natural morphism $\mathcal{F} \to \mathcal{F}^\dagger$ induced isomorphisms on stalks.*

2.1 Derived Categories

Definition 2.1.1. *The derived category on $\mathrm{Shv}(X; k)$ consists of*

i. objects complex of sheaves,

$$\mathcal{F}^\bullet := \cdots \to \mathcal{F}^{i-1} \xrightarrow{d^{i-1}} \mathcal{F}^i \xrightarrow{d^i} \mathcal{F}^{i+1}.$$

ii. The cohomology of sheaves if $\mathcal{H}(\mathcal{F}^\bullet) := \ker d^i \oplus \mathrm{im}\, d^{i-1}$.

iii. Quasi-isomorphisms $\mathcal{F}^\bullet \to \mathcal{G}^\bullet$ is called a quasi-isomorphism if it induces isomorphism in cohomologies.

The derived category = complex of sheaves / q-iso. This turns out to be a triangulated category.

However the derived category turns out to be to large to handle. Therefore, we will use a smaller category called the bounded constructible derived category of sheaves denoted by $D^b_c(X;k)$.

Definition 2.1.2. *There exists a stratification in locally closed subspaces $X = \bigsqcup_\alpha S_\alpha$ such that $\mathcal{H}^i(\mathcal{F}^\bullet)|_{S_\alpha}$ is a finite rank local system and $\mathcal{H}^i(\mathcal{F}^\bullet) = 0$ for $|i| \gg 0$.*

Six-Functors

Left Adjoints	Right Adjoints
Rf_* pushforward	f^* pullback
$f_!$	$f^!$
$(\cdot \otimes \cdot)$	$\mathrm{Hom}(\cdot,\cdot)$

Remark 2.1.1. *If f is proper then $Rf_* = f_!$, if f is open immersion then $f^* = f^!$.*

Suppose $X = U \cup V$ and $W = U \cap V$, and there are natural transformations,

$$(\cdot) \to (j_V)_*(j_U)^*(\cdot).$$

In the category of sheaves $\mathrm{Shv}(X;k) \subseteq D^b_c(X;k)$.

$$\begin{array}{ccc} \mathrm{Shv}(X;k) & \hookrightarrow & D^b(\mathrm{Shv}(X;k)) \\ \uparrow & & \uparrow \\ \mathrm{Shv}_c(X;k) & \hookrightarrow & D^b_c(\mathrm{Shv}(X;k)) \end{array}$$

Definition 2.1.3. *The Perverse sheves is $\mathrm{Perv}(X;k) \subseteq D^b_c(\mathrm{Shv}(X;k))$, where X is fnite-type variety over a field.*

Explicitly, for $D(\cdot) = \mathcal{H}om(\cdot, \omega_X)$, define $\mathcal{F} \in D^b_c(X;k)$ is a perverse sheaf if for all $n \in \mathbb{Z}$, $\dim \mathrm{supp}\, \mathcal{H}(\mathcal{F}) \le i$ and $\dim \mathrm{supp}\, \mathcal{H}(D(\mathcal{F})) \le i$.

Example 2.1.1. *Let X be a smooth variety over $\mathbb{C}$ of dimension $\dim_{\mathbb{C}} X = d$, then,*

$$\omega_X = k_X[2d],$$

then $\underline{k}_X[d]$ is a perverse sheaf and $\mathbb{D}(\underline{k}_k[d]) = \underline{k}_X[d]$. As,

$$\mathcal{H}(k_x[d]) = \begin{cases} \underline{k}_X & (i = d), \\ 0, & (\ \textit{otherwise}\ ). \end{cases}$$

Hence

$$\dim \mathrm{supp}(\mathcal{H}(\underline{k}_X[d])) = \begin{cases} d, & (i = d), \\ -\infty, & (\ \textit{otherwise}\ ). \end{cases}$$

Similarly, if $L$is a local system on X, then $L[d] \in \mathrm{Perv}(X;k)$.

Theorem 2.1.1. *$\mathrm{Perv}(X;k)$ is an Abelian category which is both Noetherian and Artinian.*

Every perverse sheaf has a composition series with ? All simple objects are constructed as follows,

$$\underbrace{X}_{\text{smooth finite-type variety}} \hookleftarrow \underbrace{Z}_{\text{closed subvariety}} \hookleftarrow \underbrace{U}_{\text{open smooth}}.$$

Let L be a local system on V, then we construct a perverse sheave,

$$\mathrm{IC}_Z(L) \subseteq \mathrm{Perv}(X;k),$$

If it is supported on Z and

$$\mathrm{IC}_Z(L)|_V = L[\dim V].$$

References

[1] Pramod Achar, Perverse Sheaves and Applications to Representation Theory.

[2] Mark de Cataldo, Luca Migliorini, The Decomposition Theorem and the topology of algebraic maps.